Code

```
% SMART FACTORY INFORMATION FLOW FOR PREDICTIVE
MAINTENANCE

clc;

clear;

close all;


%% Time

t = 1:100;


%% Simulated Sensor Data

temperature = 60 + 5*randn(1,100);     % Temperature (°C)

vibration  = 2 + 0.4*randn(1,100);     % Vibration (mm/s)


% Introduce Faults

temperature(70:100) = temperature(70:100) + 15;

vibration(80:100) = vibration(80:100) + 2;


%% Thresholds

temp_limit = 75;

vib_limit  = 3.5;


%% Fault Detection

fault = (temperature > temp_limit) | (vibration > vib_limit);


%% Predictive Maintenance Decision

maintenance = zeros(size(t));


for i = 1:length(t)

    if fault(i)
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        maintenance(i) = 1;
    end
end

%% Display Results
disp('Time  Temp  Vib  Maintenance');
disp([t' temperature' vibration' maintenance']);

%% Plot Temperature
figure;
subplot(3,1,1)
plot(t,temperature,'b','LineWidth',2)
hold on
yline(temp_limit,'r--','Threshold')
xlabel('Time')
ylabel('Temperature (°C)')
title('Machine Temperature')
grid on

%% Plot Vibration
subplot(3,1,2)
plot(t,vibration,'g','LineWidth',2)
hold on
yline(vib_limit,'r--','Threshold')
xlabel('Time')
ylabel('Vibration (mm/s)')
title('Machine Vibration')
grid on

%% Maintenance Prediction
```

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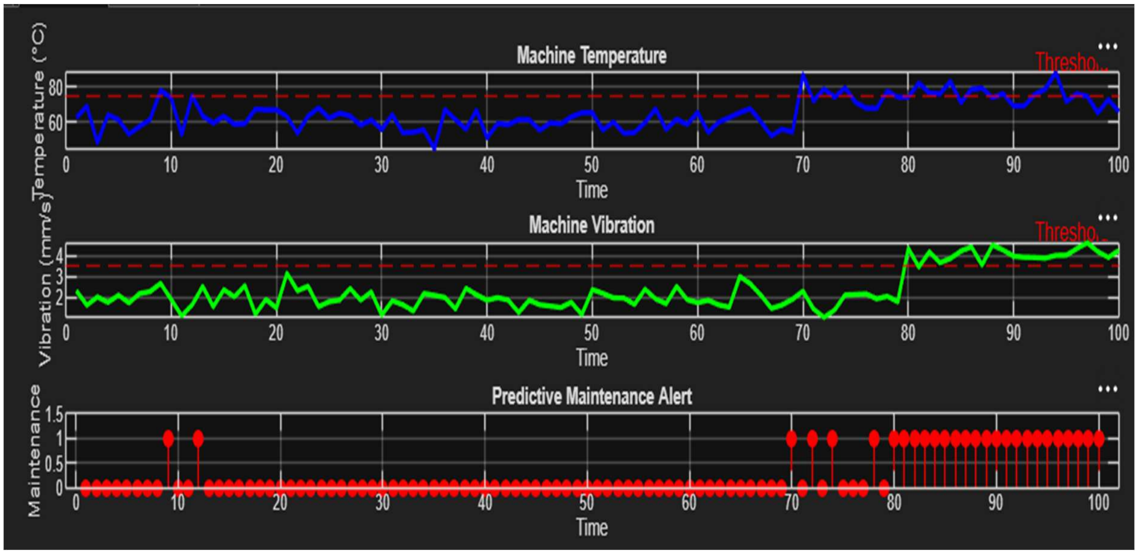
subplot(3,1,3)
stem(t,maintenance,'filled','r')
xlabel('Time')
ylabel('Maintenance')
title('Predictive Maintenance Alert')
ylim([0 1.5])
grid on

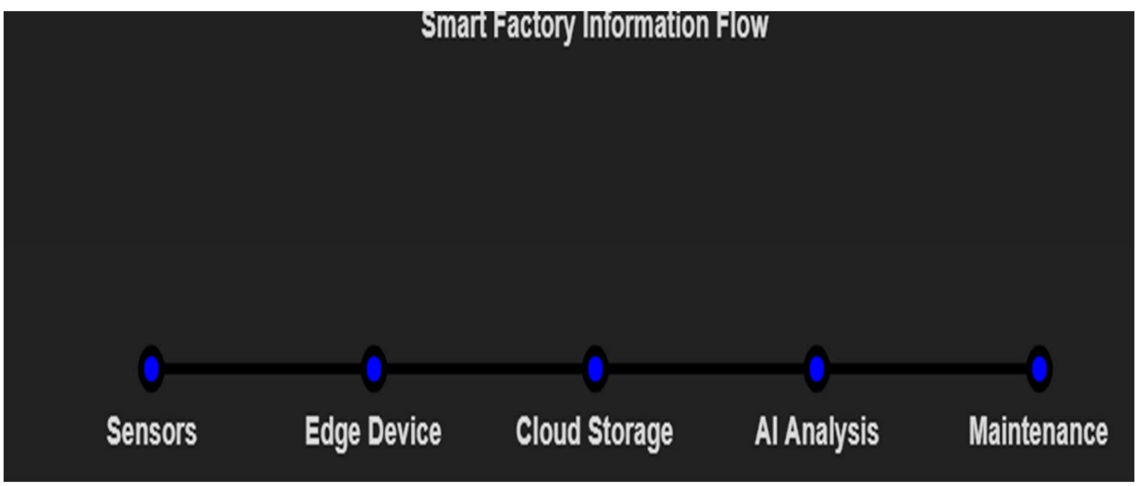
%% Information Flow
figure;
flow = {'Sensors','Edge Device','Cloud Storage','AI Analysis','Maintenance'};
x = 1:length(flow);

plot(x,ones(size(x)),'ko-','LineWidth',3,...
   'MarkerFaceColor','b','MarkerSize',10)
axis([0 6 0.5 1.5])
axis off

for i = 1:length(flow)
   text(x(i),0.9,flow{i},...
       'HorizontalAlignment','center',...
       'FontSize',11,'FontWeight','bold');
end

title('Smart Factory Information Flow')
```


Machine Temperature
Temperature (°C)
Thresho...
Time
Machine Vibration
Vibration (mm/s)
Thresho...
Time
Predictive Maintenance Alert
Maintenance
Time


Smart Factory Information Flow
Sensors
Edge Device
Cloud Storage
AI Analysis
Maintenance